IRRIGATION WATER MANAGEMENT · NOTE SP05

Irrigation Efficiencies and Land Preparation — Efficiencies, Land Levelling and Implements

Course topics: IDE-501 T5, IDE5.5 L9

Irrigation efficiency

- Irrigation efficiency expresses the performance of a whole system or its parts; it has several interpretations (e.g. application efficiency).
- **Israelsen (1950):** irrigation is the artificial application of water to soil to supply water essential for plant growth — essentially a supplement to natural precipitation. Except sub-irrigation, it is not continuous: water is applied after moisture has fallen to a level that would harm growth.
- Even with the best practice not all water stays in the root zone. **Unavoidable losses:** non-uniform distribution, waste at the ends of borders and furrows, deep percolation, evaporation from sprinkler spray and water retained on foliage. Spray evaporation and foliage retention are hard to cut; the rest can be reduced by planning, sound design and efficient operation.

$$E_i = (\text{water transpired} + \text{evaporation from soil} + \text{water for salt regulation} + \text{water used in plant tissue}) \,/\, \text{total water diverted, stored or pumped} \times 100$$

Efficiency	Formula	Meaning
Conveyance	E_c = W_f / W_r × 100	W_f = water delivered to the plot; W_r = water supplied at the source. Shows transport losses
Application	E_a = W_s / W_f × 100 = [W_f − (R_f + D_f)] / W_f × 100	W_s = water stored in the root zone; R_f = surface runoff; D_f = deep percolation (W_f = W_s + R_f + D_f, ignoring evaporation). About 60% in a well-designed surface system
Storage	E_s = W_s / W_n × 100	W_n = water needed in the root zone before irrigation (field capacity − available moisture). Shows how completely the need was met
Distribution	E_d = 100 (1 − y/d)	y = average numerical deviation in depth stored; d = average depth stored. Measures uniformity over the whole field (not per row or furrow)

- **Low application efficiency** comes from irregular land, shallow soils over permeable gravel, streams too small or too large, poor control, long runs, the wrong method, poor land preparation and excessive single applications.
- **Poor distribution** leaves dry spots, over-irrigated and waterlogged patches, and salt build-up where too little water goes — reducing yield and wasting water.

> A plot receives 100 mm, of which 10 mm runs off and 15 mm percolates below the roots: E_a = (100 − 25)/100 = 75%. If the root zone needed 80 mm, E_s = 75/80 = 94%.

Water-use efficiency

$$\text{Crop WUE} = Y / Cu; \quad \text{Field WUE} = Y / WR$$

Y = crop yield; Cu = consumptive use (ET + metabolic use); WR = total water used in the field. Crop WUE is of fundamental interest; field WUE is of greater practical importance.

Preparing land for irrigation

Efficient surface irrigation needs a smooth surface and uniform distribution; land shaping controls the flow, reduces erosion and improves drainage. Land preparation has four stages:

Stage	Work
Land clearing	Remove boulders, stones, stumps, unwanted trees and brush
Rough grading	Remove high spots (mounds, ridges) and fill pockets, gullies and low areas; usually followed by levelling, smoothing or both
Land levelling	Reshape the surface to a planned grade, usually one or a series of planes per field
Land smoothing (planing)	Remove minor irregularities without changing the general topography; the final operation

- **Levelling limits:** topsoil depth limits how far you can level, especially in shallow soils. Yields may dip for a while where subsoil is mixed into the root zone — restore productivity with fertiliser and good management. Levelling may not pay where deep cuts meet unresponsive subsoil. Use animal-drawn buck scrapers or tractor-drawn carry-all scrapers; animals cannot handle long hauls.
- **Smoothing:** the highs and lows are small and hard to see, the cuts and fills light. A land plane or wooden float shaves the highs into the lows in three or more passes — crosswise, lengthwise and a final pass in the direction of irrigation.
- **Planning:** level field by field, but study the whole farm first; divide it into areas of similar slope and soil. Survey by scattered spot levels or a grid; contour maps show the profile and the extent of cuts and fills. Plan field boundaries, irrigation and drainage channels and farm roads before levelling.

Rainfall and land slope

Rainfall sets the permissible grades: the minimum must meet drainage needs; the maximum must not produce erosive velocities.

Soil (Table 8.5)	Recommended safe slope (%)
Heavy (clay) soils	0.10 – 0.30
Medium (loam) soils	0.20 – 0.35
Light (sandy) soils	0.25 – 0.60

Border irrigation allows no cross slope and needs precise down-field grading; furrow irrigation tolerates some cross slope and less precise grades.

Contour bench levelling

- Prepares steep or undulating land — where irrigation is difficult or hazardous — by dividing it into strips along approximate contours, each levelled independently; it also controls erosion from rain.
- **Section:** strips up to 10 m wide are common; vertical interval preferably ≤ 30 cm and never over 60 cm; bunds high enough to hold the irrigation stream and storm runoff.
- **Layout:** keep benches close to the original topography (a topographic map helps but is not essential on smooth slopes); no cross slope; longitudinal slope within Table 8.5; avoid sharp bends; run benches continuously across the field; integrate drains.

- **Construction:** scrape across the strips at right angles, moving soil from the upper part to the lower; form the ridges on the uphill return trip; finish with a float.

Implements for land preparation

Implement	Use
Buck scraper	The most useful low-cost animal-drawn levelling implement; moves loosened soil (loosened by mould-board plough, disc harrow or desi plough) from high to low areas; pulled by an average pair of bullocks (shorten to 1.5 m for smaller bullocks)
Bulldozer	Crawler tractor with a dozer blade; cuts and pushes earth short distances (≈ 25 m); for rough grading and heavy earthwork; not economical for small farm jobs
Tractor-drawn scrapers	From tracer blades to heavy carrier scrapers; land grading; not generally for small farms
Wooden float	The most suitable animal-drawn smoothing implement; float lengthwise, crosswise and diagonally (3 or more passes); 1.0 – 1.8 m wide for animals, at least 5 m long for precision; float fields before every seeding
Wheeled scrapers and leveller blades	Small-scale levelling: scrape ploughed soil, drag it a short distance and dump it; smooth with tractor land planes or animal floats
V-ditcher	Builds and cleans irrigation and drainage channels: survey and stake the line, open a furrow with a mould-board plough, then run the ditcher back and forth for a symmetrical ditch; the yoke must be long enough for the animals to walk outside the channel
A-frame ridger	Home-made; makes ridges and bunds; the operator rides on it — the nearer he stands to the bullocks, the bigger the ridge
Disc ridger	Tractor-operated, two gangs of two discs facing each other to gather loosened soil; ideal for ridges and bunds on prepared fields

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) lessons on soil–water–plant relations, evapotranspiration, irrigation efficiencies, and surface, sprinkler and drip irrigation. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Water application efficiency of a well-designed surface system is about

(a) 30%
(b) 60%
(c) 85%
(d) 95%

Q2. Water storage efficiency compares water stored with

(a) water delivered
(b) water needed in the root zone
(c) water diverted
(d) rainfall

Q3. Distribution efficiency E_d equals

(a) 100(1 - y/d)
(b) 100 y/d
(c) d/y
(d) W_s/W_f

Q4. Recommended safe slope for heavy clay soils is

(a) 0.10–0.30%
(b) 0.25–0.60%
(c) 1–2%
(d) 3–5%

Q5. The vertical interval of contour benches should preferably not exceed

(a) 10 cm (b) 30 cm

(c) 1 m (d) 2 m

Q6. The most suitable animal-drawn implement for land smoothing is the

(a) V-ditcher (b) wooden float

(c) disc ridger (d) bulldozer

Answers

Q1 — (b) 60%

Q2 — (b) water needed in the root zone

Q3 — (a) 100(1 - y/d)

Q4 — (a) 0.10–0.30%

Q5 — (b) 30 cm — *Never more than 60 cm.*

Q6 — (b) wooden float